From Statistics to Estimators

In our previous discussions, we established that:

- A **sample** $X_1, \ldots, X_n$ is drawn from a population with an unknown parameter θ.
- A **statistic** $T(X)$ is any function of the sample data (e.g., $\bar{X}$, M_n).
- Because the sample is random, every statistic is a **random variable** governed by a **sampling distribution**.

When we use a statistic specifically to guess the value of an unknown population parameter θ, we promote it to the title of an **estimator**, denoted as $\hat{\theta}$.

Since $\hat{\theta}$ is a random variable, it will almost never perfectly equal θ in any single sample. So, what mathematical properties define a "good" estimator?

Desirable Properties: Unbiasedness (Accuracy)

Because an estimator $\hat{\theta}$ is a random variable, it will sometimes overestimate and sometimes underestimate θ.

Unbiasedness: *Is our estimator correct on average?*

An estimator $\hat{\theta}$ is called an **unbiased estimator** of θ if its expected value equals the true parameter:

$$\mathbb{E}[\hat{\theta}] = \theta \quad \text{for all possible values of } \theta$$

The **bias** of an estimator is defined as the difference between its expected value and the true parameter:

$$\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$$

- If $\text{Bias}(\hat{\theta}) = 0$, the estimator is perfectly unbiased.
- *Recall:* We previously saw that $\mathbb{E}[\bar{X}] = \mu$, proving the sample mean is an unbiased estimator of the population mean.

Desirable Properties: Efficiency (Precision)

Unbiasedness ensures we are right on average, but we also want our estimates to be reliably close to the true value in almost every sample.

Efficiency relates to the **variance** of the estimator, $\mathrm{Var}(\hat{\theta})$.

- A smaller variance means the estimator's sampling distribution is tightly concentrated around its expected value.
- Given two unbiased estimators, $\hat{\theta}_1$ and $\hat{\theta}_2$, the one with the smaller variance is considered more **efficient**.

Example: If data is drawn from a Normal distribution, both the sample mean and the sample median are unbiased estimators of the true center μ. However, the sample mean has a notably smaller variance, making it the more efficient choice.

The Bias-Variance Tradeoff: Mean Squared Error

How do we evaluate an estimator that might have a slight bias but a very small variance? We use the **Mean Squared Error (MSE)**.

The MSE measures the average squared difference between the estimator and the true parameter:

$$\text{MSE}(\hat{\theta}) = \mathbb{E}\left[(\hat{\theta} - \theta)^2\right]$$

By expanding the mathematical expectation, MSE neatly decomposes into two components:

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + \left[\text{Bias}(\hat{\theta})\right]^2$$

- A robust estimator minimizes the MSE.
- In modern statistics, we are sometimes willing to accept a tiny amount of bias if it drastically reduces the variance, leading to a superior overall estimator.

Proof: MSE Decomposition

Let $\mu = \mathbb{E}[\hat{\theta}]$.:

$$\begin{aligned}
\text{MSE}(\hat{\theta}) &= \mathbb{E}\left[(\hat{\theta} - \theta)^2\right] \\
&= \mathbb{E}\left[(\hat{\theta} - \mu + \mu - \theta)^2\right] \\
&= \mathbb{E}\left[(\hat{\theta} - \mu)^2 + 2(\hat{\theta} - \mu)(\mu - \theta) + (\mu - \theta)^2\right] \\
&= \mathbb{E}\left[(\hat{\theta} - \mu)^2\right] + 2(\mu - \theta)\mathbb{E}\left[\hat{\theta} - \mu\right] + (\mu - \theta)^2
\end{aligned}$$

Since ($\mathbb{E}[\hat{\theta} - \mu] = 0$), the middle term vanishes:

$$\begin{aligned}
\text{MSE}(\hat{\theta}) &= \text{Var}(\hat{\theta}) + 0 + \left[\text{Bias}(\hat{\theta})\right]^2 \\
&= \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2
\end{aligned}$$

Desirable Properties: Consistency

Finally, we want to know what happens to our estimator as we collect more data (as $n \to \infty$).

An estimator $\hat{\theta}_n$ (where n denotes sample size) is **consistent** if it converges in probability to the true parameter θ.

Informally, as the sample size grows infinitely large, the probability that our estimate is off by any arbitrary margin shrinks to zero:

$$\lim_{n \to \infty} \mathbb{P}(|\hat{\theta}_n - \theta| > \epsilon) = 0 \quad \text{for any } \epsilon > 0$$

- **Connection:** Just as the Central Limit Theorem explains the *shape* of large samples, the Law of Large Numbers guarantees that $\bar{X}$ is a *consistent* estimator of μ. More data forces the estimate closer to the truth.

Types of Estimators: Point vs. Interval

Once we have chosen a "good" estimator (unbiased, efficient, consistent), we must decide how to report our guess about the population parameter θ.

There are two primary paradigms in classical inference:

1. **Point Estimation:** Provides a single "best guess" value for the unknown parameter.
   - *Output:* A single number (e.g., $\hat{\mu} = 42.5$).
2. **Interval Estimation:** Provides a range of plausible values for the parameter, quantifying the uncertainty of our guess.
   - *Output:* An interval (e.g., $40.1 \leq \mu \leq 44.9$).

Point Estimators

A **point estimator** uses the sample data to compute a single statistic that serves as the "best" estimate of the population parameter.

Common Examples:

- Sample mean ($\bar{X}$) estimates the population mean (μ).
- Sample variance (S^2) estimates the population variance (σ^2).
- Sample proportion ($\hat{p}$) estimates the population probability of success (p).

The Limitation: While a point estimator like $\bar{X}$ might be unbiased and minimize MSE, it provides no information about its own accuracy in a specific sample. We know it fluctuates due to sampling variation, but a single number doesn't tell us *how much* it might be off by.

Interval Estimators (Confidence Intervals)

To address the limitations of point estimation, we use **interval estimators**. Instead of a single point, we construct lower and upper bounds (L, U) from the sample data.

The most common form is the **Confidence Interval (CI)**.

- We associate the interval with a **confidence level**, typically denoted as $(1 - \alpha)$, where α is often 0.05.
- A 95% confidence interval implies that if we repeated our sampling process indefinitely, 95% of the constructed intervals would contain the true parameter θ.
- *Crucial Interpretation:* The parameter θ is fixed; it is the interval bounds L and U that are random variables varying from sample to sample.

Interval Estimators

For many standard models (often relying on the Central Limit Theorem), a confidence interval takes a symmetric form:

$$\textbf{Point Estimate} \pm \textbf{Margin of Error}$$

Example: CI for μ (known σ)
Using the Standard Normal distribution (Z), the $100(1-\alpha)\%$ confidence interval is:

$$\bar{X} \pm Z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)$$

Constructing Point Estimators

We have discussed the properties of a "good" estimator (unbiasedness, efficiency, consistency), but how do we actually come up with the mathematical formula for $\hat{\theta}$ in the first place?

Rather than guessing and checking properties, classical statistics relies on systematic frameworks to construct estimators from data.

The two most foundational methods are:

1. **Method of Moments (MoM):** An intuitive, algebraic approach based on matching sample statistics to theoretical properties.
2. **Maximum Likelihood Estimation (MLE):** A probabilistic and calculus-based approach that seeks the parameter value that makes the observed data most probable.

Method of Moments (MoM):

Method of Moments relies on a very simple and intuitive principle: **Equate what you observe in the sample to what you expect in the population.**

- **Population Moments:** The theoretical expected values, which are functions of the unknown parameters $\theta_1, \ldots, \theta_k$.

$$\mu_j = \mathbb{E}[X^j]$$

- **Sample Moments:** The empirical averages computed directly from our observed data $X_1, \ldots, X_n$.

$$m_j = \frac{1}{n} \sum_{i=1}^{n} X_i^j$$

Method of Moments (MoM):

If a model has k unknown parameters $\theta_1, \ldots, \theta_k$, we need k equations to solve for them.

1. Calculate the first k theoretical moments $\mu_1, \mu_2, \ldots, \mu_k$ as functions of $\theta_1, \ldots, \theta_k$.
2. Set up a system of k equations by equating them to the sample moments:

$$\mu_1(\theta_1, \ldots, \theta_k) = \frac{1}{n}\sum X_i$$
$$\mu_2(\theta_1, \ldots, \theta_k) = \frac{1}{n}\sum X_i^2$$
$$\vdots$$
$$\mu_k(\theta_1, \ldots, \theta_k) = \frac{1}{n}\sum X_i^k$$

3. Solve the system algebraically for $\theta_1, \ldots, \theta_k$. The solutions are your MoM estimators: $\tilde{\theta}_1, \ldots, \tilde{\theta}_k$.

Theoretical Properties of MoM Estimators

Let $\theta = g(\mu_1, \ldots, \mu_k)$ be the parameter expressed as a function of the population moments.

1. **Consistency (Asymptotic Correctness):**
   - By the Law of Large Numbers (LLN), sample moments converge in probability to population moments ($m_j \xrightarrow{p} \mu_j$).
   - If $g(\cdot)$ is a continuous function, the **Continuous Mapping Theorem** ensures:

$$\tilde{\theta}_{MoM} = g(m_1, \ldots, m_k) \xrightarrow{p} g(\mu_1, \ldots, \mu_k) = \theta$$

2. **Bias (Small Sample Error):**
   - The expectation operator is strictly linear. If $g(\cdot)$ is a **non-linear** function (like $1/x$, x^2, or $\sqrt{x}$):

$$\mathbb{E}[\tilde{\theta}_{MoM}] = \mathbb{E}[g(m_1, \ldots, m_k)] \neq g(\mathbb{E}[m_1, \ldots, m_k]) = \theta$$

   - Thus, whenever the parameter is a non-linear function of the moments, the MoM estimator will be biased in finite samples.

MoM Example 1: Poisson Distribution

Suppose $X_1, \ldots, X_n$ are i.i.d. observations from a Poisson distribution with unknown parameter λ.

1. Identify the parameter and required moments:

- Parameter to estimate: $k = 1$ (λ).
- We need only the first moment (1 equation).

2. Calculate population and sample moments:

- Theoretical first moment: $\mu_1 = \mathbb{E}[X] = \lambda$.
- Sample first moment: $m_1 = \frac{1}{n}\sum_{i=1}^{n} X_i = \bar{X}$.

3. Equate and solve:

$$\mu_1 = m_1 \implies \lambda = \bar{X}$$

Thus, the MoM estimator for the Poisson distribution is simply the sample mean: $\tilde{\lambda}_{MoM} = \bar{X}$.

MoM Example 2: Uniform Distribution

Suppose $X_1, \ldots, X_n$ are i.i.d. from a Uniform distribution on $[0, \theta]$ with unknown maximum boundary θ.

1. Identify moments:

- Theoretical first moment: $\mu_1 = \mathbb{E}[X] = \frac{0+\theta}{2} = \frac{\theta}{2}$.
- Sample first moment: $m_1 = \bar{X}$.

2. Equate and solve:

$$\begin{aligned} \mu_1 &= m_1 \\ \frac{\theta}{2} &= \bar{X} \\ \theta &= 2\bar{X} \end{aligned}$$

Thus, the MoM estimator is $\tilde{\theta}_{MoM} = 2\bar{X}$.

Intuition: If the average observed value in your sample is 5, the Method of Moments guesses that the maximum possible value (θ) is 10.

MoM Example 3: Normal Distribution

Suppose $X_1, \ldots, X_n \sim \mathcal{N}(\mu, \sigma^2)$. We have $k = 2$ parameters (μ and σ^2), so we need two moments.

1. **Theoretical Moments:**
   - $\mu_1 = \mathbb{E}[X] = \mu$
   - $\mu_2 = \mathbb{E}[X^2] = \mathrm{Var}(X) + (\mathbb{E}[X])^2 = \sigma^2 + \mu^2$
2. **Equate to Sample Moments (m_1, m_2):**

$$\mu = m_1$$
$$\sigma^2 + \mu^2 = m_2$$

3. **Solve for parameters:** Substituting the first equation into the second:

$$\tilde{\mu} = m_1 = \bar{X}$$
$$\tilde{\sigma}^2 = m_2 - m_1^2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2 - \bar{X}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

MoM Example 4: Gamma Distribution

Suppose $X_1, \ldots, X_N \sim \text{Gamma}(n, \lambda)$ with shape n and rate λ. *(Note: We use N for sample size here to avoid confusion with the shape parameter n).*

1. Theoretical vs. Sample Moments: *Shortcut: We can use the sample variance ($s^2 = m_2 - m_1^2$) to simplify the algebra.*

- Mean: $\mathbb{E}[X] = \frac{n}{\lambda} = \bar{X}$
- Variance: $\text{Var}(X) = \frac{n}{\lambda^2} = s^2$

2. Solve for n and λ: Divide the mean equation by the variance equation to isolate λ:

$$\frac{n/\lambda}{n/\lambda^2} = \frac{\bar{X}}{s^2} \implies \tilde{\lambda} = \frac{\bar{X}}{s^2}$$

Substitute $\tilde{\lambda}$ back into the mean equation:

$$\frac{n}{\tilde{\lambda}} = \bar{X} \implies n = \tilde{\lambda}\bar{X} \implies \tilde{n} = \frac{\bar{X}^2}{s^2}$$

MoM Example 5: Binomial Distribution (Discrete)

Suppose $X_1, \ldots, X_N \sim \text{Binomial}(n, p)$ with unknown number of trials n and success probability p.

1. Theoretical vs. Sample Moments: Using the sample mean ($\bar{X}$) and sample variance (s^2) shortcut again:

- Mean: $\mathbb{E}[X] = np = \bar{X}$
- Variance: $\text{Var}(X) = np(1-p) = s^2$

2. Solve for n and p: Substitute the mean equation ($np = \bar{X}$) directly into the variance equation:

$$\bar{X}(1-p) = s^2$$

$$1 - p = \frac{s^2}{\bar{X}} \implies \tilde{p} = 1 - \frac{s^2}{\bar{X}} = \frac{\bar{X} - s^2}{\bar{X}}$$

Substitute $\tilde{p}$ back into the mean equation to find n:

$$n\left(\frac{\bar{X} - s^2}{\bar{X}}\right) = \bar{X} \implies \tilde{n} = \frac{\bar{X}^2}{\bar{X} - s^2}$$

Why don't we use these Binomial estimates?

The estimators $\tilde{p} = 1 - \frac{s^2}{\bar{X}}$ and $\tilde{n} = \frac{\bar{X}^2}{\bar{X} - s^2}$ perfectly illustrate the **limitations of the Method of Moments**.

In practice, these are rarely used for three reasons:

- **Parameter Space Violations:** n must be an integer, but $\tilde{n}$ will almost certainly be a fraction.
- **Nonsensical Results:** For a binomial, variance must be strictly less than the mean. If random sampling noise causes our sample variance to exceed the sample mean ($s^2 > \bar{X}$), then $\tilde{p}$ and $\tilde{n}$ become *negative*.
- **Context:** In almost all real-world experiments, the number of trials (n) is known beforehand, making it a 1-parameter problem where $\tilde{p} = \bar{X}/n$.

MoM Limitations: Uniform Distribution Support

Recall our MoM estimator for $X \sim \text{Uniform}(0, \theta)$:

$$\tilde{\theta}_{MoM} = 2\bar{X}$$

- By definition, every data point must be less than or equal to the maximum boundary: $X_i \leq \theta$ for all i.
- This means the true θ must be at least as large as the maximum observed value: $\theta \geq \max(X_1, \ldots, X_N)$.

A Problematic Sample:

- Suppose we observe the sample: $1, 2, 9$.
- The sample mean is $\bar{X} = 4$.
- The MoM estimate is $\tilde{\theta}_{MoM} = 2(4) = 8$.

Conclusion: The MoM estimate claims the maximum possible value is 8, but we literally observed a 9! MoM ignores the hard boundaries (support) of the distribution.

MoM Limitations: The Cauchy Distribution

What happens if the theoretical moments do not exist? Consider the **Cauchy distribution** with location parameter θ:

Probability Density Function (PDF):

$$f(x;\theta) = \frac{1}{\pi\left[1 + (x-\theta)^2\right]}, \quad -\infty < x < \infty$$

(Think of it as a bell curve with extremely "heavy tails".)

The Breakdown of MoM:

- To use MoM, we need to calculate the first theoretical moment, $\mathbb{E}[X]$.
- However, because the tails of the Cauchy distribution are so thick, the integral $\int_{-\infty}^{\infty} xf(x)dx$ does not converge.
- **The mean is mathematically undefined.**

Conclusion: If a distribution has undefined or infinite theoretical moments, the Method of Moments completely fails. You cannot equate a sample mean to an undefined theoretical value.

Maximum Likelihood Estimation (MLE)

While MoM is algebraically simple, **Maximum Likelihood Estimation** is the workhorse of modern parametric statistics due to its strong asymptotic properties (efficiency and normality).

The Paradigm Shift:

- *Probability* asks: Given a known parameter θ, what data are we likely to observe?
- *Likelihood* asks: Given the specific data we *already* observed, what parameter value θ makes this data the most probable?

We want to find the parameter $\hat{\theta}_{MLE}$ that maximizes the joint probability (or density) of the observed sample.

The Likelihood and Log-Likelihood Functions

Let $X_1, \ldots, X_n$ be an i.i.d. sample from a distribution with pdf or pmf $f(x \mid \theta)$.

The **Likelihood Function**, $L(\theta)$, is the joint density evaluated at the observed data, treated as a function of θ:

$$L(\theta \mid \mathbf{x}) = \prod_{i=1}^{n} f(x_i \mid \theta)$$

Because products are difficult to differentiate and can be very close to zero when dealing with large data, we almost always maximize the natural logarithm, known as the **Log-Likelihood**, $\ell(\theta)$:

$$\ell(\theta) = \ln L(\theta) = \sum_{i=1}^{n} \ln f(x_i \mid \theta)$$

Since the natural logarithm is a strictly increasing function, the θ that maximizes $\ell(\theta)$ is exactly the same θ that maximizes $L(\theta)$.

Finding the Maximum Likelihood Estimator

To find the MLE, we use standard calculus optimization techniques on the log-likelihood function.

The Procedure:

1. Formulate the log-likelihood function $\ell(\theta)$.
2. Take the first derivative with respect to θ (this derivative is known as the **score function**).
3. Set the derivative equal to zero to find the critical points:

$$\frac{\partial}{\partial \theta}\ell(\theta) = 0$$

4. Solve for θ. This solution is $\hat{\theta}_{MLE}$.
5. Verify it is a maximum by checking the second derivative: $\frac{\partial^2}{\partial \theta^2}\ell(\theta) < 0$.

MLE Example 1: Poisson Distribution

Let $X_1, \ldots, X_n \sim \text{Poisson}(\lambda)$. The pmf is $f(x \mid \lambda) = \frac{e^{-\lambda}\lambda^x}{x!}$.

Step 1: The Likelihood Function

$$L(\lambda) = \prod_{i=1}^{n} \frac{e^{-\lambda}\lambda^{x_i}}{x_i!} = \frac{e^{-n\lambda}\lambda^{\sum x_i}}{\prod x_i!}$$

Step 2: The Log-Likelihood

$$\ell(\lambda) = \ln L(\lambda) = -n\lambda + \left(\sum_{i=1}^{n} x_i\right) \ln(\lambda) - \ln\left(\prod_{i=1}^{n} x_i!\right)$$

Step 3 & 4: Differentiate and Solve

$$\frac{\partial}{\partial\lambda}\ell(\lambda) = -n + \frac{\sum x_i}{\lambda}$$

Setting the derivative to zero and solving for λ:

$$n = \frac{\sum x_i}{\lambda} \implies \hat{\lambda}_{MLE} = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{X}$$

MLE Example 2: Exponential Distribution

Let $X_1, \ldots, X_n \sim \text{Exponential}(\lambda)$, where λ is the rate parameter. The pdf is $f(x \mid \lambda) = \lambda e^{-\lambda x}$ for $x \geq 0$.

Step 1: The Likelihood Function

$$L(\lambda) = \prod_{i=1}^{n} \lambda e^{-\lambda x_i} = \lambda^n e^{-\lambda \sum x_i}$$

Step 2: The Log-Likelihood: $\ell(\lambda) = \ln L(\lambda) = n \ln(\lambda) - \lambda \sum_{i=1}^{n} x_i$

Step 3 & 4: Differentiate and Solve

$$\frac{\partial}{\partial \lambda} \ell(\lambda) = \frac{n}{\lambda} - \sum_{i=1}^{n} x_i = 0$$

$$\frac{n}{\lambda} = \sum_{i=1}^{n} x_i \implies \hat{\lambda}_{MLE} = \frac{n}{\sum x_i} = \frac{1}{\bar{X}}$$

MLE Example 3: Discrete Uniform Distribution

Let $X_1, \ldots, X_n \sim$ Discrete Uniform$(1, N)$. The pmf is $f(x \mid N) = \frac{1}{N}$ for $x \in \{1, 2, \ldots, N\}$.

Step 1: The Likelihood Function

$$L(N) = \prod_{i=1}^{n} \frac{1}{N} I(x_i \leq N) = \frac{1}{N^n} I(N \geq x_{(n)})$$

where $x_{(n)} = \max(x_1, \ldots, x_n)$ and I is the indicator function.

Step 2: Maximization (No Calculus Needed) The likelihood $L(N) = \frac{1}{N^n}$ is a strictly decreasing function of N. To maximize it, we must choose the smallest possible integer value for N. The indicator function requires $N \geq x_{(n)}$, so:

$$\hat{N}_{MLE} = \max(X_1, \ldots, X_n) = X_{(n)}$$

MLE Example 3: Discrete Uniform Distribution

Contrast with Method of Moments (MoM) Equating population mean to sample mean:

$$E[X] = \frac{N+1}{2} = \bar{X} \implies \hat{N}_{MoM} = 2\bar{X} - 1$$

MLE vs. MoM: MoM can yield illogical estimates (e.g., non-integers, or an upper limit $\hat{N}_{MoM} < x_{(n)}$). The MLE is logically consistent as it strictly guarantees the estimate bounds all observed data.

MLE Example 4: Normal Distribution

Let $X_1, \ldots, X_n \sim \mathrm{N}(\mu, \sigma^2)$. The pdf is $f(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$.

Step 1: The Likelihood Function

$$L(\mu, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i -$$

Step 2: The Log-Likelihood

$$\ell(\mu, \sigma^2) = \ln L(\mu, \sigma^2) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2$$

MLE Example 4: Normal Distribution

Step 3: Differentiate w.r.t μ

$$\frac{\partial}{\partial \mu}\ell(\mu, \sigma^2) = -\frac{1}{2\sigma^2}\sum_{i=1}^{n} 2(x_i - \mu)(-1) = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i - \mu)$$

Step 4: Solve for μ Setting the derivative to zero:

$$\frac{1}{\sigma^2}\left(\sum_{i=1}^{n} x_i - n\mu\right) = 0$$

$$n\mu = \sum_{i=1}^{n} x_i \implies \hat{\mu}_{MLE} = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{X}$$

MLE Example 4: Normal Distribution

Step 5: Differentiate w.r.t σ^2 Let $v = \sigma^2$. We differentiate $\ell(\mu, v)$ with respect to v:

$$\frac{\partial}{\partial v}\ell(\mu, v) = -\frac{n}{2v} + \frac{1}{2v^2}\sum_{i=1}^{n}(x_i - \mu)^2$$

Step 6: Solve for σ^2 Setting the derivative to zero and substituting $\hat{\mu}_{MLE} = \bar{X}$:

$$\frac{n}{2v} = \frac{1}{2v^2}\sum_{i=1}^{n}(x_i - \bar{X})^2 \implies n = \frac{1}{v}\sum_{i=1}^{n}(x_i - \bar{X})^2$$

$$\hat{\sigma}^2_{MLE} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{X})^2$$

MLE and MoM: A Frequent Coincidence

Observation from Previous Examples

- For the Poisson distribution, both MLE and MoM yield $\hat{\lambda} = \bar{X}$.
- For the Normal distribution, MoM equates $E[X] = \bar{X}$ and $E[X^2] = \frac{1}{n}\sum x_i^2$, yielding $\hat{\mu} = \bar{X}$ and $\hat{\sigma}^2 = \frac{1}{n}\sum(x_i - \bar{X})^2$. These are exactly the MLEs.

Is this a coincidence? No, there is a structural reason why these two distinct philosophical approaches often produce identical estimators.

The General Criteria:

The Setup Consider a probability distribution of the form:

$$f(x \mid \boldsymbol{\theta}) = \exp\left(a_0 + a_1 x + a_2 x^2 + \dots\right)$$

The Log-Likelihood For a sample of size n, the log-likelihood is:

$$\ell(\boldsymbol{\theta}) = b_0 + b_1 x + b_x^2 + b_3 x^3 + \dots$$

WHere a_i and b_is depend on $\theta = (\theta_1, \theta_2, \theta_3, \dots)$.
In these specific cases the MoM estimates and MLE estimates coincide.

Example: Continuous Uniform $(0, \theta)$ MLE $\neq$ MoM

Let $X_1, \ldots, X_n \sim \text{Uniform}(0, \theta)$. The pdf is $f(x \mid \theta) = \frac{1}{\theta}$ for $0 < x < \theta$.

Method of Moments (MoM) Equating the first population moment to the sample moment:

$$E[X] = \frac{\theta}{2} = \bar{X} \implies \hat{\theta}_{MoM} = 2\bar{X}$$

Maximum Likelihood Estimate (MLE) The likelihood is $L(\theta) = \frac{1}{\theta^n} I(X_{(n)} \leq \theta)$. To maximize $L(\theta)$, we minimize θ subject to the indicator constraint:

$$\hat{\theta}_{MLE} = X_{(n)} = \max(X_1, \ldots, X_n)$$

Variance Comparison: Uniform $(0, \theta)$

Let's compare the variances of these two distinct estimators.

Variance of MoM: Since $\text{Var}(X) = \frac{\theta^2}{12}$, the variance of the sample mean is $\text{Var}(\bar{X}) = \frac{\theta^2}{12n}$.

$$\text{Var}(\hat{\theta}_{MoM}) = \text{Var}(2\bar{X}) = 4\left(\frac{\theta^2}{12n}\right) = \frac{\boldsymbol{\theta^2}}{\mathbf{3n}} \quad \sim \mathcal{O}(n^{-1})$$

Variance of MLE: Using the pdf of the maximum order statistic $f(x_{(n)}) = \frac{nx^{n-1}}{\theta^n}$:

$$\text{Var}(\hat{\theta}_{MLE}) = \frac{\mathbf{n}}{(\mathbf{n+2})(\mathbf{n+1})^{\mathbf{2}}}\theta^{\mathbf{2}} \quad \sim \mathcal{O}(n^{-2})$$

Conclusion: For $n \geq 2$, $\text{Var}(\hat{\theta}_{MLE}) < \text{Var}(\hat{\theta}_{MoM})$. The MLE variance shrinks much faster $(1/n^2)$ than the MoM variance $(1/n)$.

General Rule: Asymptotic Efficiency

Why does MLE usually beat MoM?
While finite-sample variances (like the Uniform example) can sometimes be complex to compare directly, the advantage of MLE becomes a mathematical guarantee in large samples.

Under standard regularity conditions, Maximum Likelihood Estimators are **asymptotically efficient**. As $n \to \infty$:

$$\text{Var}(\hat{\theta}_{MLE}) \approx \frac{1}{I_n(\theta)}$$

where $I_n(\theta)$ is the Fisher Information.

The MLE attains the Cramér-Rao Lower Bound (CRLB). Since MoM estimators are generally not efficient unless they coincide with the MLE, we conclude:

$$\text{Var}(\hat{\theta}_{MLE}) \leq \text{Var}(\hat{\theta}_{MoM})$$

Functions of Parameters: The Invariance Theorem

Theorem (Invariance Property) Let $\hat{\theta}$ be the Maximum Likelihood Estimator of a parameter θ. If $\eta = g(\theta)$ is a one-to-one function of θ, then the Maximum Likelihood Estimator of η is given by:

$$\hat{\eta}_{MLE} = g(\hat{\theta}_{MLE})$$

Two Ways to Find $\hat{\eta}_{MLE}$ When we want to find the MLE of a transformed parameter $\eta = g(\theta)$, we can mathematically take one of two paths:

1. **Direct Maximization:** Reparameterize the likelihood function strictly in terms of η (using $\theta = g^{-1}(\eta)$), and differentiate to find the maximum.
2. **Invariance Theorem:** Find the MLE of the original parameter θ, and simply plug it into the function $g(\theta)$.

Example: Exponential Distribution (Method 1: Direct)

Let $X_1, \ldots, X_n \sim \text{Exp}(\lambda)$ with pdf $f(x \mid \lambda) = \lambda e^{-\lambda x}$. We want to find the MLE for the mean, defined as $\eta = g(\lambda) = \frac{1}{\lambda}$.

Method 1: Direct Reparameterization Substitute $\lambda = \frac{1}{\eta}$ into the pdf: $f(x \mid \eta) = \frac{1}{\eta} e^{-x/\eta}$.
The log-likelihood becomes:

$$\ell(\eta) = \ln\left(\prod_{i=1}^{n} \frac{1}{\eta} e^{-x_i/\eta}\right) = -n\ln(\eta) - \frac{1}{\eta}\sum_{i=1}^{n} x_i$$

Differentiate w.r.t η and set to zero:

$$\frac{\partial \ell}{\partial \eta} = -\frac{n}{\eta} + \frac{1}{\eta^2}\sum_{i=1}^{n} x_i = 0$$

$$\frac{n}{\eta} = \frac{\sum x_i}{\eta^2} \implies \hat{\eta}_{MLE} = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{\mathbf{x}}$$

Example: Exponential Distribution (Method 2: Invariance)

Method 2: Using the Invariance Theorem First, find the MLE for the original parameter λ using the standard log-likelihood $\ell(\lambda) = n\ln(\lambda) - \lambda\sum x_i$:

$$\frac{\partial \ell}{\partial \lambda} = \frac{n}{\lambda} - \sum_{i=1}^{n} x_i = 0 \implies \hat{\lambda}_{MLE} = \frac{n}{\sum x_i} = \frac{1}{\bar{X}}$$

Now, apply the Invariance Theorem. Since $\eta = g(\lambda) = \frac{1}{\lambda}$, we just plug in $\hat{\lambda}_{MLE}$:

$$\hat{\eta}_{MLE} = g(\hat{\lambda}_{MLE}) = \frac{1}{\hat{\lambda}_{MLE}} = \frac{1}{\left(\frac{1}{\bar{X}}\right)} = \bar{\mathbf{X}}$$

Conclusion: Both paths lead to the exact same estimator ($\bar{X}$). The Invariance Theorem allows us to bypass the algebra of reparameterizing and re-differentiating the likelihood function!

Invariance Example: Poisson Probabilities

The Setup Let $X_1, \ldots, X_n \sim \text{Poisson}(\lambda)$. From our first example, we know the MLE for the rate parameter is the sample mean:

$$\hat{\lambda}_{MLE} = \bar{X}$$

The Problem Suppose a researcher doesn't want the rate λ, but specifically wants to estimate the probability that **zero events** occur in a given interval. Mathematically, this is:

$$\eta = g(\lambda) = P(X = 0) = e^{-\lambda}$$

Invariance Example: Applying the Theorem

Step 1: Verify the function is one-to-one The function $g(\lambda) = e^{-\lambda}$ is strictly decreasing for all $\lambda > 0$. Because it is strictly monotonic, it is a one-to-one function.

Step 2: Apply the Invariance Theorem We do not need to rewrite the likelihood function $L(\eta)$ and differentiate from scratch. We simply apply the transformation to the known MLE of λ:

$$\hat{\eta}_{MLE} = g(\hat{\lambda}_{MLE}) = \mathbf{e}^{-\bar{\mathbf{x}}}$$

Asymptotic Properties: Consistency of MLE

Theorem (Consistency of MLE) Under standard regularity conditions (e.g., identifiable parameters, true parameter in the interior of the parameter space, and the log-likelihood is sufficiently smooth/differentiable), the Maximum Likelihood Estimator is consistent:

$$\hat{\theta}_{MLE} \xrightarrow{p} \theta_0 \text{ i.e } \lim_{n\to\infty} P(|\hat{\theta}_n - \theta_0| > \epsilon) = 0$$

In simple terms: with enough data, the MLE will perfectly locate the true parameter.

Consistency in Our Previous Examples

The WLLN Connection For our earlier examples in the exponential family, the MLEs were directly tied to sample moments. By the Weak Law of Large Numbers (WLLN) and the Continuous Mapping Theorem, consistency is easily verified:

- **Poisson:** $\hat{\lambda}_{MLE} = \bar{X}$. By WLLN, $\bar{X} \xrightarrow{p} E[X] = \lambda$.
- **Normal:** $\hat{\mu}_{MLE} = \bar{X} \xrightarrow{p} \mu$.
  $\hat{\sigma}^2_{MLE} = \frac{1}{n}\sum(X_i - \bar{X})^2 \xrightarrow{p} \text{Var}(X) = \sigma^2$.
  (Even though $\hat{\sigma}^2$ is biased for finite n, the bias $\frac{1}{n}\sigma^2 \to 0$, so it is still consistent.)
- **Exponential:** $\hat{\lambda}_{MLE} = \frac{1}{\bar{X}}$. Since $\bar{X} \xrightarrow{p} \frac{1}{\lambda}$, continuous mapping yields $\frac{1}{\bar{X}} \xrightarrow{p} \lambda$.

Consistency Beyond Regular Conditions

The Uniform Distribution Case Recall our Uniform $(0, \theta)$ example, where the support of the distribution depends on the parameter θ.

Does the MLE consistency theorem fail here?

No. The MLE is $\hat{\theta}_{MLE} = X_{(n)}$. The CDF of $X_{(n)}$ is $F_{X_{(n)}}(x) = \left(\frac{x}{\theta}\right)^n$ for $0 < x < \theta$. For any $\epsilon > 0$:

$$P(|X_{(n)} - \theta| \geq \epsilon) = P(X_{(n)} \leq \theta - \epsilon) = \left(\frac{\theta - \epsilon}{\theta}\right)^n$$

As $n \to \infty$, this probability goes to 0. Thus, $X_{(n)} \xrightarrow{p} \theta$.